

despite the inherent randomness in the solute distribution, demonstrates the model’s ability to reliably predict the Peierls barrier across varying atomic environments. This result highlights the model’s capacity to grasp complex atomic interactions that influence the Peierls barrier.

From these predictions, we can compute the mean energy barrier for each composition, a key parameter in solute strengthening theories, which rely on these barriers to predict material strength. Figure 2(b) compares the mean Peierls barrier values predicted by the DL model against the ground truth values, showing a strong agreement across both ternary and binary alloys. This alignment highlights the model’s ability to predict individual configurations while also capturing broader trends in solute-strengthening behavior, offering valuable insights into how solute distributions influence material properties across various compositions.

In summary, the model’s low error and consistent performance across diverse alloy compositions affirm its potential as a powerful tool for predicting material behavior, significantly reducing the need for computationally expensive atomistic simulations such as NEB calculations.

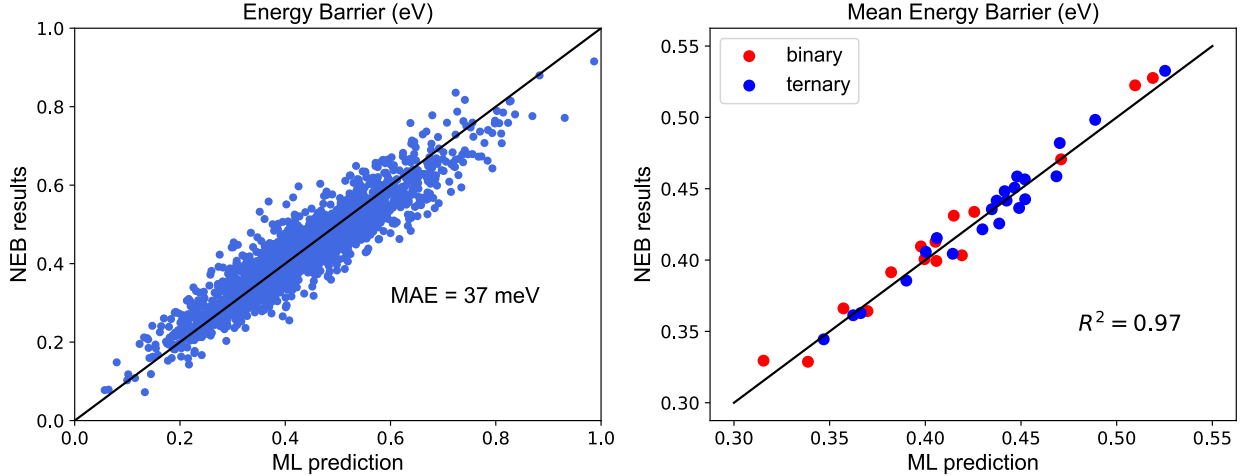


Figure 2: **Evaluation of the GNN model for Peierls barrier prediction on the test set.** Test set contains new compositions that never appeared in the training set. (a) Comparison of machine learning (ML) predictions and NEB results for the Peierls energy barrier. (b) Comparison of the ML results and NEB results for the mean energy barriers of binary and ternary compositions in the test set. The correlation coefficient R^2 between the predictions and the ground truth is 0.97.

2.1.2 Potential energy change predictions

Another important quantity is the potential energy change as the dislocation moves from one energy minimum to the next. After training the GNN using the potential energy changes as graph labels, we compared the model’s predictions with the ground truth values for the graphs in the test set (see Fig. S1), as shown in Figure 3(a). The relatively low mean absolute error of 60 meV demonstrates the strong performance of our DL model in accurately predicting potential energy changes.

The random fluctuations in the solute environment lead to a distribution of energy changes. In random alloys, the average of this distribution is expected to be zero, and according to solute strengthening theories, the energy barrier against dislocation motion scales with the standard deviation of this distribution [26, 24]. This can be expressed as:

$$\Delta \tilde{E}_p(a) = \left(\frac{b}{\zeta} \right)^{\frac{1}{2}} \sigma_{\Delta U} \quad (1)$$

where ζ is the dislocation segment length (with $\zeta = 5b$ in our case), and b is the screw dislocation Burgers vector.

Figure 3(b) shows the values of the solute/screw interaction energy parameter computed by our DL model compared to the ground truth values, demonstrating good agreement for both ternary and binary alloys.